

API Documentation

Simplified Network Monitoring System API v1

Base URL: `https://api.example.com/api/v1`

Version: 1.0

Date: January 19, 2026

Authentication

JWT Bearer Token

Login:

```
POST /api/v1/auth/login
Content-Type: application/json

{
  "username": "admin",
  "password": "securePass123"
}
```

Response:

```
{
  "success": true,
  "data": {
    "accessToken": "eyJhbGc...",
    "refreshToken": "eyJhbGc...",
    "expiresIn": 900,
    "user": {
      "id": "uuid",
      "username": "admin",
      "role": "administrator"
    }
  }
}
```

Using Token:

```
Authorization: Bearer eyJhbGc...
```

API Key

```
X-API-Key: sk_live_abc123
```

Response Format

```
{
  "success": boolean,
  "data": T | null,
```

```
"error": {
  "code": string,
  "message": string
} | null,
"meta": {
  "timestamp": string,
  "requestId": string,
  "pagination": {...} | null
}
```

Pagination & Filtering

```
GET /devices?page=2&pageSize=20&filter[status]=active&sort=-created_at
```

Devices API

GET /devices

List all devices.

Response:

```
{
  "data": [
    {
      "id": "uuid",
      "name": "Server-01",
      "type": "server",
      "ipAddress": "192.168.1.100",
      "status": "active",
      "sensorCount": 12
    }
  ]
}
```

POST /devices

Create device.

Request:

```
{
  "name": "Server-02",
  "type": "server",
  "ipAddress": "192.168.1.101"
}
```

Sensors API

POST /sensors

Create sensor.

Request:

```
{
  "deviceId": "uuid",
  "name": "CPU Usage",
  "type": "snmp",
  "interval": 60,
  "config": {
    "oid": "1.3.6.1.4.1.2021.11.9.0"
  }
}
```

Metrics API

GET /metrics/query

Query time-series data.

Parameters:

- `deviceId` (required)
- `from` (ISO8601, required)
- `to` (ISO8601, required)
- `aggregation` : avg, min, max
- `interval` : 1m, 5m, 1h, 1d

Response:

```
{
  "data": {
    "series": [
      {
        "sensorName": "CPU Usage",
        "dataPoints": [
          { "timestamp": "2026-01-19T00:00:00Z", "value": 42.5 }
        ]
      }
    ]
  }
}
```

Alerts API

GET /alerts

List alerts.

Response:

```
{
  "data": [
    {
      "id": "uuid",
      "severity": "warning",
      "status": "triggered",
      "message": "CPU usage 85%",
      "deviceName": "Server-01",
      "triggeredAt": "2026-01-19T11:30:00Z"
    }
  ]
}
```

```
]
}
```

POST /alerts/:id/acknowledge

```
{"comment": "Investigating"}
```

Dashboards API

POST /dashboards

Create dashboard.

Request:

```
{
  "name": "Network Overview",
  "widgets": [
    {
      "type": "chart",
      "config": {"sensorId": "uuid"},
      "position": {"x": 0, "y": 0, "w": 6, "h": 3}
    }
  ]
}
```

WebSocket API

Connection

```
io('wss://api.example.com', {
  auth: { token: 'jwt...' }
});
```

Events

Client → Server:

```
socket.emit('subscribe:device', 'device-uuid');
```

Server → Client:

```
socket.on('metric:update', (data) => {
  // Real-time metric updates
});

socket.on('alert:triggered', (alert) => {
  // New alerts
});
```

Error Codes

Code	Status	Description
UNAUTHORIZED	401	Auth required
FORBIDDEN	403	No permission
RESOURCE_NOT_FOUND	404	Not found
VALIDATION_ERROR	400	Invalid input
RATE_LIMIT_EXCEEDED	429	Too many requests

Rate Limits

- **Authenticated:** 1000 req/hour
- **API Keys:** 5000 req/hour
- **WebSocket:** 10 concurrent connections

Headers:

```
X-RateLimit-Limit: 1000
X-RateLimit-Remaining: 999
X-RateLimit-Reset: 1642598400
```